

Sustainable Removal of PFAS

Dr. Zhiming Zhang and Dr. Kauser Jahan

As emerging contaminants, per- and polyfluoroalkyl substances (PFAS) are drawing growing concerns due to their persistence in the environment and potential for acute health impacts. PFAS compounds are frequently detected in urban stormwater runoff, groundwater, and even tap water. This project will investigate the removal of PFAS via filtration and adsorption processes.

- Types and concentrations of PFAS in tap water will be explored, followed by the treatment by household filters.
- Different types of ion-exchange resins will be tested for the removal of PFAS compounds via batch sorption tests in the lab.
- Regeneration of these ion-exchange resins will be performed to achieve optimal utilization conditions.

